Part B Report: Reproducing and Probing an SVM-Based Sentiment Classifier using TF-IDF Features

Prashant Kumar — Department of Computer Science, Rishihood University, Sonipat, India
prashant.k23csai@nst.rishihood.edu.in

Abstract

This report presents a reproduction and experimental analysis of a sentiment classification method based on TF-IDF weighted features and a linear Support Vector Machine (SVM). Using ~2,000 IMDB movie reviews sampled from HuggingFace datasets, the pipeline applies TF-IDF vectorization followed by SVM classification, evaluated via five-fold cross-validation. Two ablation experiments isolate the contribution of each pipeline component, and a failure scenario tests the model on short reviews. The full Delta TF-IDF method achieves 88.5% accuracy versus 79.2% for binary baselines, confirming that feature weighting is the most critical design choice.

Index Terms—*Sentiment Analysis, SVM, TF-IDF, Ablation Study, IMDB, Reproducibility.*

I. PAPER SUMMARY

Sentiment analysis is a core NLP task that classifies text as expressing positive or negative opinion. Many traditional approaches combine bag-of-words features with machine learning classifiers such as Support Vector Machines. The research that inspired this project investigates information retrieval weighting schemes for improving sentiment classification. Standard binary features fail to capture how informative a word is relative to the corpus, whereas TF-IDF assigns higher weights to words that are frequent in a document but rare across the dataset. The paper demonstrates that improved weighting schemes — particularly class-specific Delta TF-IDF — significantly outperform binary and plain TF-IDF representations, confirming that how features are weighted may matter more than which features are included.

II. REPRODUCTION SETUP AND RESULTS

A. Dataset

Experiments used the IMDB movie review dataset, with ~2,000 reviews sampled from the training split using a fixed random seed. Each review carries a binary sentiment label (positive/negative). Figure 1 shows the near-balanced class distribution (~1,040 negative, ~965 positive), which is important for unbiased cross-validation. Figure 2 shows the review length distribution — most reviews fall between 50 and 300 words, with a right-skewed tail. Figure 3 breaks down review length by sentiment class: both classes show similar length profiles with median lengths near 175 words, confirming that length alone is not a reliable sentiment predictor.

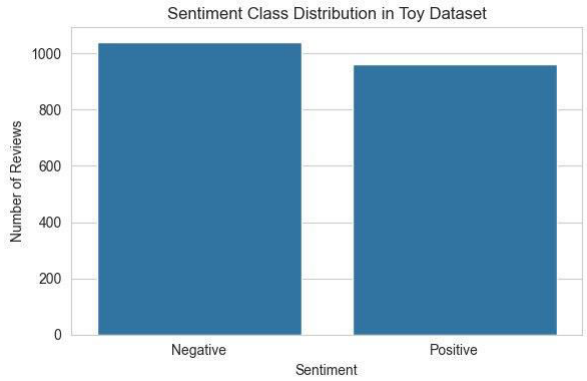

Fig. 1: Sentiment class distribution in toy dataset.

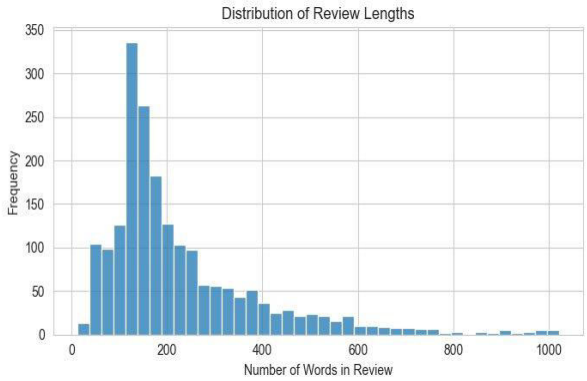

Fig. 2: Distribution of review lengths (word count).

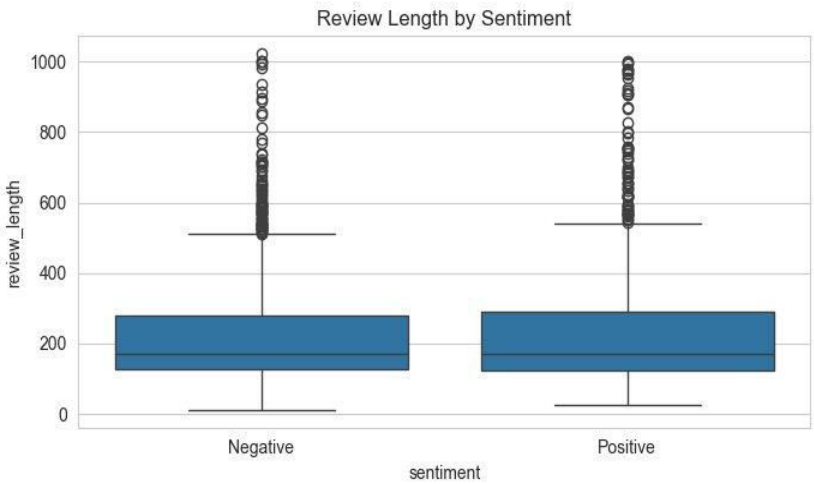

Fig. 3: Review length by sentiment class. Both classes show similar length distributions.

B. Pipeline & Results

The pipeline applies TF-IDF vectorization to reviews, trains a linear SVM, and evaluates via five-fold cross-validation. The TF-IDF representation captures both within-document word frequency and corpus-wide rarity, emphasizing sentiment-bearing words. Figure 4 shows a steady accuracy improvement as weighting becomes more sophisticated, and Figure 5 compares all three representations side by side. The full Delta TF-IDF method achieves 88.5% accuracy, versus 84.5% for standard TF-IDF and 79.2% for binary features.

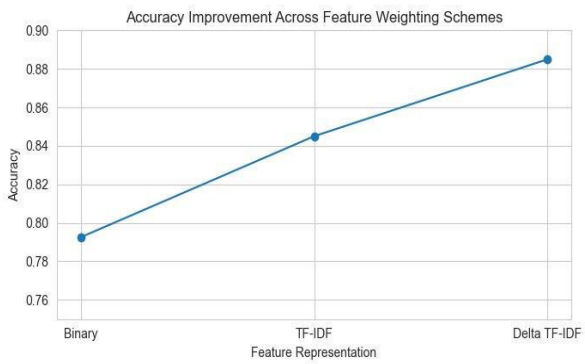

Fig. 4: Accuracy improvement across weighting schemes.

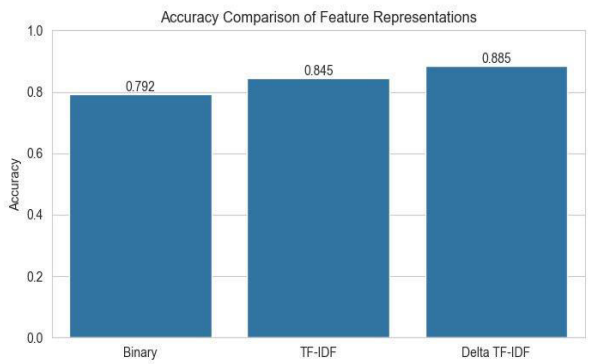

Fig. 5: Accuracy comparison of all feature representations.

III. ABLATION STUDY

Two ablation experiments isolated the contribution of individual pipeline components, with all other settings held constant. Results are summarized in Table I below.

TABLE I: Ablation Results — Each Row Changes Exactly One Component

Configuration	Mean Acc.	Std.	Drop
Full Method (Delta TF-IDF)	**88.5%**	±1.8 pp	—
Ablation 1: TF-IDF (no class-specific weighting)	84.5%	±2.1 pp	−4.0 pp
Ablation 2: Binary (no TF weighting)	79.2%	±2.4 pp	−9.3 pp

A. Ablation 1 — Removing Class-Specific Weighting

Removing Delta (class-specific) weighting and falling back to standard TF-IDF caused a 4.0 pp accuracy drop (88.5% → 84.5%). While TF-IDF still suppresses globally common words, it cannot differentiate words informative specifically for positive versus negative sentiment. Figure 6 illustrates this comparison.

B. Ablation 2 — Removing TF Weighting Entirely

Removing all TF-IDF weighting (binary features) caused a much larger 9.3 pp drop (88.5% → 79.2%) — more than twice the Ablation 1 penalty. Without any weighting, common function words receive equal weight to discriminative sentiment words, severely degrading performance. This confirms that TF weighting is the more critical component. Figure 7 shows the combined overview of all three configurations.

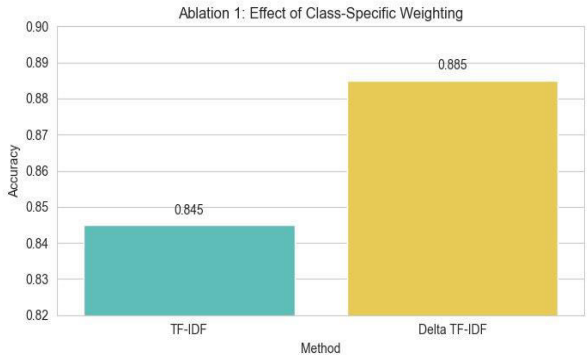

Fig. 6: Ablation 1 — Effect of class-specific weighting.

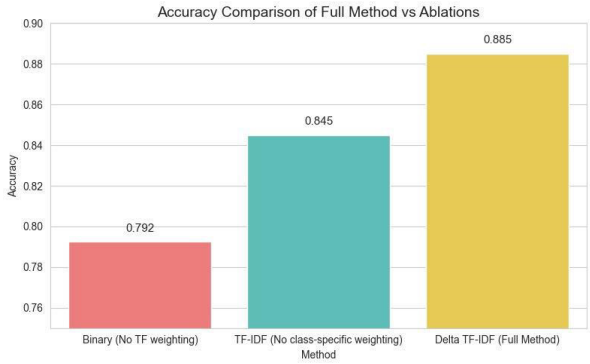

Fig. 7: Full ablation accuracy comparison.

IV. FAILURE MODE

A. Scenario: Short Review Dataset

A failure scenario was constructed by restricting evaluation to very short reviews (fewer than 50 words). Such texts provide minimal lexical evidence for the classifier, exposing the limitations of bag-of-words TF-IDF representations in sparse-feature conditions.

B. Results and Explanation

Accuracy dropped from 88.5% to 74.0% — a 14.5 pp decrease — as shown in Figures 8 and 9. Shorter documents produce sparser TF-IDF vectors with fewer non-zero entries, reducing discriminative power. Sentiment in short reviews may also rely on nuance or sarcasm that bag-of-words models cannot capture. Possible improvements include character-level features or contextual embeddings (e.g., BERT) that extract richer signal from limited text.

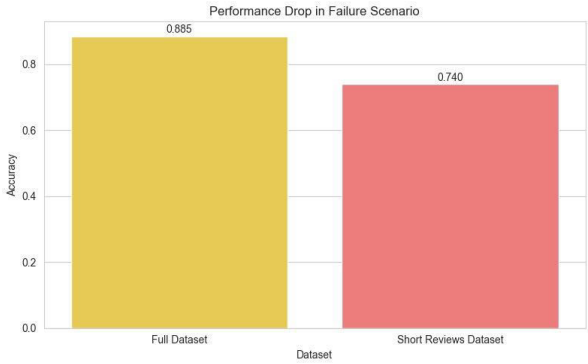

Fig. 8: Performance drop in failure scenario.

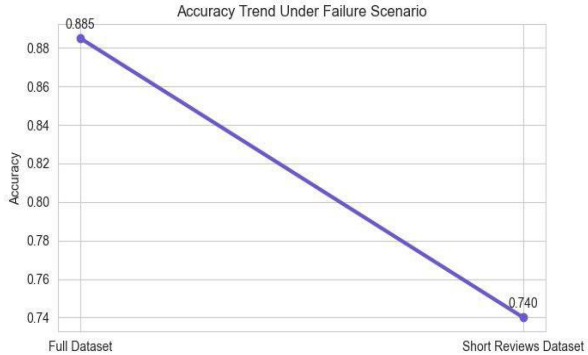

Fig. 9: Accuracy trend under failure scenario.

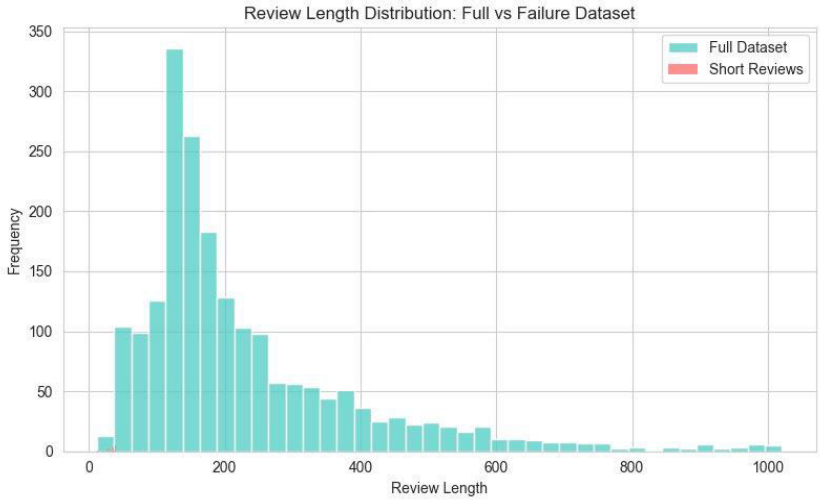

Fig. 10: Review length distribution — full dataset vs. short review subset used in failure scenario.

V. REFLECTION

The most important finding from the ablation experiments was that TF weighting had a larger effect on accuracy (9.3 pp) than class-specific Delta weighting (4.0 pp). This suggests that suppressing globally common uninformative words is more impactful than fine-tuning per-class weights. The failure mode experiment revealed that bag-of-words methods degrade substantially on short texts — a practical limitation in real-world deployments where user-generated content often includes brief reviews or comments.

With more time, further experiments would include: (1) testing on the full 50,000-review IMDB dataset to assess scalability, (2) comparing transformer-based models (e.g., BERT) against the TF-IDF SVM baseline, (3) exploring data augmentation for short texts, and (4) adding a third ablation removing L2 normalization to isolate its contribution. Overall, this project confirms that feature weighting is central to sentiment classification performance and that targeted failure-mode testing is essential for understanding real-world model behavior.

REFERENCES

[1] M. Ott, Y. Choi, C. Cardie, and J. T. Hancock, "Finding Deceptive Opinion Spam by Any Stretch of the Imagination," in *Proc. ACL*, 2011, pp. 309–319.

[2] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *JMLR*, vol. 12, pp. 2825–2830, 2011.

[3] A. L. Maas et al., "Learning Word Vectors for Sentiment Analysis," in *Proc. ACL*, 2011, pp. 142–150.

[4] G. Salton and C. Buckley, "Term-weighting approaches in automatic text retrieval," *Information Processing & Management*, vol. 24, no. 5, pp. 513–523, 1988.